MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3VACECT200GreenElectronics

(MG3VACECC200-*Doublemajorcoursecode)*

Duration:1.5hours **MaximumMarks:50**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I)andAppreciation(Ap)*

Part A

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries1mark

1	WhichofthefollowingisNOTacommonsourceofE-waste? A) Discardedmobilephones B)Oldtelevision sets C)Worn-outclothes D)Brokencomputerparts	U	1
2	In E-waste classification, devices like printers, scanners, and fax machines fall under which category? A) Largehouseholdappliances B)Consumerelectronics C)SmallITandtelecommunicationequipmentD)Medical devices	U	1
3	WhichharmfuleffectcanE-wastehaveonaquaticecosystems? A) Increaseindissolvedoxygen B) Releaseoftoxicheavymetalsintowater bodies C) Growthofcoralreefs D) Removalofoilspills	U	1
4	WhichcategoryofE-wasteincludesrefrigerators,washingmachines,andair conditioners? A) Smallhouseholdappliances B)Largehouseholdappliances C)Lightingequipment D)Electricaltools	U	1
5	WhichheavymetalinE-wasteisknowntocausedamagetothenervoussystem? A) Copper B)Lead C)Aluminum D) Nickel	U	1
6	OnemajorchallengeinE-wastedisposalis: A) Lackofenoughlandfillspace B)Lowcostofrecycling C)Abundanceoftrainedworkers D)Excessivedemandforrecycledgoods	U	1
7	WhichofthefollowingisabenefitofE-wasterecycling? A) Increasedelectronicimports B) Recoveryofvaluablemetalslikegoldandsilver C) Increasedlandfillpollution D) Reductioninemploymentopportunities	C	3
8	ExposuretomercuryfromE-wastecanmainlyaffectthe: A) Respiratorysystem B)Digestivesystem C)Nervous system D)Skeletalsystem	U	1
9	WhichtoxiccompoundfoundinE-wasteiscommonlyusedinsoldering materials? A) Mercury B)Lead C)Cadmium D)Chromium	U	1
10	Cadmium,presentinrechargeablebatteriesinE-waste,primarilyaffects the: A) Skin B)Kidneys C)Heart D)Bones	U	1
11	Youaretaskedwithdesigninganeco-friendlycircuitboard.Whichtoxic compound should you avoid to prevent nervous system damage? A) Nickel B) Copper C)Mercury D)Aluminum	C	3
12	An electronics startup wants to create a corrosion-resistant coating without using a harmful E-waste compound. Which one should be excluded from the design?	C	3

	A) ChromiumVI B)Nickel C)Copper D)Tin		
13	Sustainablematerialsinelectronicsaremainlychosenbecausethey: A) Increasedevice size B)Havenoneedforrecycling C)Reduceenvironmentalimpact D)Makedevices heavier	A	2
14	Whichofthefollowingisnotafeatureofsustainablematerialsin electronics? A)Abilitytobe recycled B)Hightoxicity C)Reducedresourcedepletion D)Longerlifespaninuse	A	2
15	Theadoptionofsustainablematerialsinelectronicsprimarilyhelpsto: A)Increasetheamountofe-waste B)Conservenaturalresources C)Shortenproductlifecycles D)Increaseminingoperations	A	2
16	Usingbiodegradablecasingsforelectronicssupportswhichenvironmental approach? A)Planned obsolescence B)Circulareconomy C)Overproduction D)Electronicisolation	A	2
17	WhichmetalismostcommonlyrecoveredfromPCBsforusein wiring? A)Aluminum B)Gold C)Copper D)Nickel	A	2
18	Whencomparingperformance,certainsustainableelectronicmaterialscan provide: A)Equalconductivitywithaddedbiodegradability B)Highertoxicitywithlowercost C)Fasterwear-outratesinallapplications D)Nopossibilityofintegrationintodevices	A	2
19	Themainreasongoldisusedinelectronicconnectorsis: A)Itischeaperthancopper B)Itresistscorrosionandconductselectricitywell C)Itmeltsatlowtemperatures D)Itislighterthanaluminum	A	2
20	Heatsinksinelectronicdevicesareoftenmadefrom recycled: A)Gold B)Aluminum C) Silver D)Lead	A	2
21	TheprocessofrecyclingcopperfromPCBshelps reduce: A)Device weight B)Greenhousegasemissions C)Electricalconductivity D)Toxicityof gold	A	2
22	Comparedtotraditionalmaterials,sustainable materials areoften chosenbecause they: A)IncreaseminingdemandB)Haveunlimitedlifespanregardlessofusage C)Reduceresourcedepletionandenvironmentalharm D)Arenon-recyclable	A	2
23	GoldrecoveryfromPCBsismoreefficient when: A)ComponentsaredismantledbeforeprocessingB)PCBsareburneddirectly C)PCBsarecrushedwithout sorting D)Goldisreplacedwith copper	A	2
24	Whicheco-friendlydielectricmaterialisderivedfromplantfibersandoffersgood biodegradability? A) Resin B)Cellulose C)Shellac D)Gelatin	A	2
25	Magneticseparationinrecyclingisspecificallyusedtorecover: A)Non-ferrousmetalslikealuminumB)Ferrousmetalslikeironandsteel C)Plasticsfrompaper D)Glassfrommetals	C	3
26	Airclassificationinmechanicalrecyclingseparatesmaterialsbased on: A)Electricalconductivity B)Magneticproperties C)Weightandaerodynamicbehavior D)Thermalresistance	C	3
27	Inmechanicalrecycling,theshreddingprocessisprimarilyusedto: A)Sortmaterialsby density B)Breakdownwasteintosmallerpiecesforeasierhandling C)Removemagneticmetalsfromnon-metals D)Separatelightmaterialsfromheavyones	C	3
28	Gravityseparationismosteffectivefor: A)Separatingmaterialsbasedondensitydifferences B)Sortingmaterialsbycolor C)Melting plastics	C	3

	D)ExtractinggoldfromPCBs		
29	Inmaterialrecovery,leachingismainlyusedto: A)Dissolvevaluablemetalsfromsolidwasteusingsuitablechemical solutions B)Burnwasteathightemperatureswithout oxygen C)Separatemetalsusingmagneticforces D)Crushmaterialsintofineparticles	C	3
30	Pyrolysisisachemicalprocessthat: A)Usesheatintheabsenceofoxygentodecomposeorganicmaterials B)Separatesmetalsbymagneticproperties C)Electroplatesmetalsontonewsurfaces D)Dissolvesmetalsinacidicsolutions	C	3
31	Electrochemicalprocessesinmaterialrecoveryarecommonlyusedfor: A)Extractingandrefiningmetalsthroughcontrolledelectriccurrent B)Separatingplasticsfrommetalsusingstaticelectricity C)Removingoxygenfrommolten metal D)Burningwastetogenerateheat	C	3
32	InPCdismantling,thepowersupplyunit(SMPS)canbereusedmainlyfor: A)Storing data B)ConvertingACtoDCpower C)Displayingimages D)Coolingtheprocessor	A	4
33	TheheatsinkinsideaPCismainlyusedfor: A)Storingsoftware B)ReducingCPUtemperature C)IncreasingnetworkspeedD)Storingelectrical charge	A	4
34	WhichreusablePC component isresponsiblefor temporarydatastorageduring processing? A) Harddiskdrive B)RAM C)Powersupplyunit D)Heatsink	A	4
35	WhichofthefollowingisNOTareusablecomponentinsideaPC? A)Harddiskdrive B)RAMmodule C)Motherboard D)Burnt-outfuse	A	4

1 * 35 = 35

Part B

Short Answer Questions
Answer any 3
QuestionsEachquestioncarrie
s5marks

36	Writeashortnoteonthesourcesandhealthimpactsofmercury,lead,and cadmium found in E-waste.	U	1
37	Listanyfiveeco-friendlydielectricmaterialsandmentiononepropertyorusefor each	A	2
38	Explainanyfourmechanicalrecyclingmethodsusedinmaterialrecovery,with one example for each.	C	3
39	Listanyfivereusablecomponentsthatcanberecoveredfromapersonal computer. How each recovered part can be repurposed?	A	4
40	ShortnoteonE-wastecompoundsanditstoxicitychart	C	3

3 * 5 = 15